

SUBROUTINE PARAMETERS AND DATA LINKAGE

Linkage of Parameters and Data between the Main Program and a Subroutine

- via Registers
- via Memory locations
-

Example: Subroutine performing *LOGICAL OR operation*; Need two parameters

Loc.			
		ORG 200	
200		LDA X	/ Load 1st operand into AC
201		BSA OR	/ Branch to subroutine OR
202		HEX 3AF6	/ 2nd operand stored here
203		STA Y	/ Subroutine returns here
204		HLT	
205	X,	HEX 7B95	/ 1st operand stored here
206	Y,	HEX 0	/ Result stored here
207	OR,	HEX 0	/ Subroutine OR
208		CMA	/ Complement 1st operand
209		STA TMP	/ Store in temporary location
20A		LDA OR I	/ Load 2nd operand
20B		CMA	/ Complement 2nd operand
20C		AND TMP	/ AND complemented 1st operand
20D		CMA	/ Complement again to get OR
20E		ISZ OR	/ Increment return address
20F		BUN OR I	/ Return to main program
210	TMP,	HEX 0	/ Temporary storage
		END	



SUBROUTINE - Moving a Block of Data -

		/ Main program
	BSA MVE	/ Branch to subroutine
	HEX 100	/ 1st address of source data
	HEX 200	/ 1st address of destination data
	DEC -16	/ Number of items to move
	HLT	
MVE,	HEX 0	/ Subroutine MVE
	LDA MVE I	/ Bring address of source
	STA PT1	/ Store in 1st pointer
	ISZ MVE	/ Increment return address
	LDA MVE I	/ Bring address of destination
	STA PT2	/ Store in 2nd pointer
	ISZ MVE	/ Increment return address
	LDA MVE I	/ Bring number of items
	STA CTR	/ Store in counter
	ISZ MVE	/ Increment return address
LOP,	LDA PT1 I	/ Load source item
	STA PT2 I	/ Store in destination
	ISZ PT1	/ Increment source pointer
	ISZ PT2	/ Increment destination pointer
	ISZ CTR	/ Increment counter
	BUN LOP	/ Repeat 16 times
	BUN MVE I	/ Return to main program
PT1,	--	
PT2,	--	
CTR,	--	



INPUT OUTPUT PROGRAM

Program to Input one Character(Byte)

CIF,	SKI	/ Check input flag
	BUN CIF	/ Flag=0, branch to check again
	INP	/ Flag=1, input character
	OUT	/ Display to ensure correctness
	STA CHR	/ Store character
	HLT	
CHR,	--	/ Store character here

Program to Output a Character

	LDA CHR	/ Load character into AC
COF,	SKO	/ Check output flag
	BUN COF	/ Flag=0, branch to check again
	OUT	/ Flag=1, output character
	HLT	
CHR,	HEX 0057	/ Character is "W"

